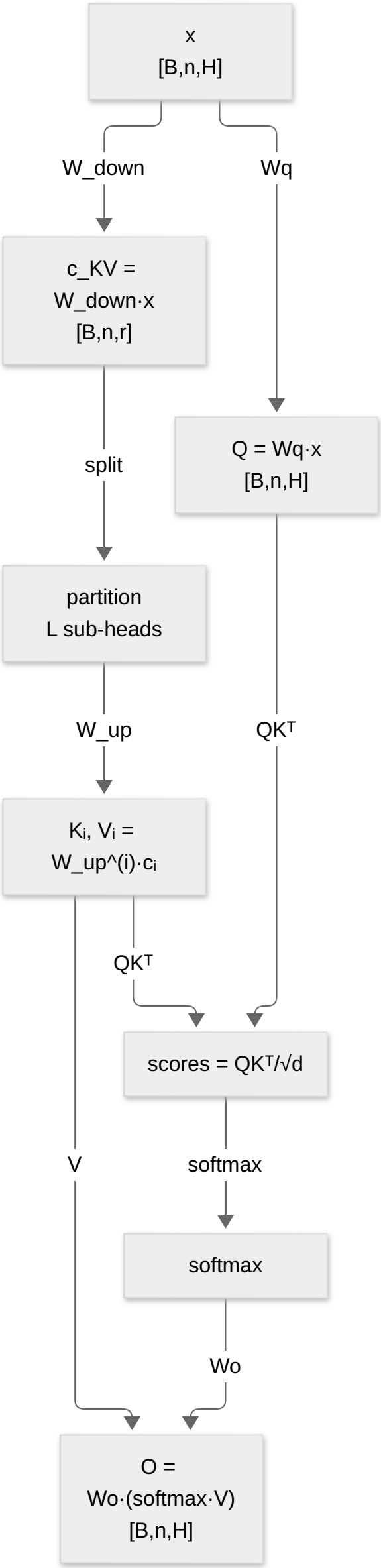

x
[B,n,H]
W_down
Wq
c_KV =
W_down·x
[B,n,r]
split
Q = Wq·x
[B,n,H]
partition
L sub-heads
W_up
QKᵀ
Kᵢ, Vᵢ =
W_up^(i)·cᵢ
QKᵀ
scores = QKᵀ/√d
V
softmax
softmax
Wo
O =
Wo·(softmax·V)
[B,n,H]